🧪 Lab 2: Secure & Scalable VLAN Design (Router-on-a-Stick)

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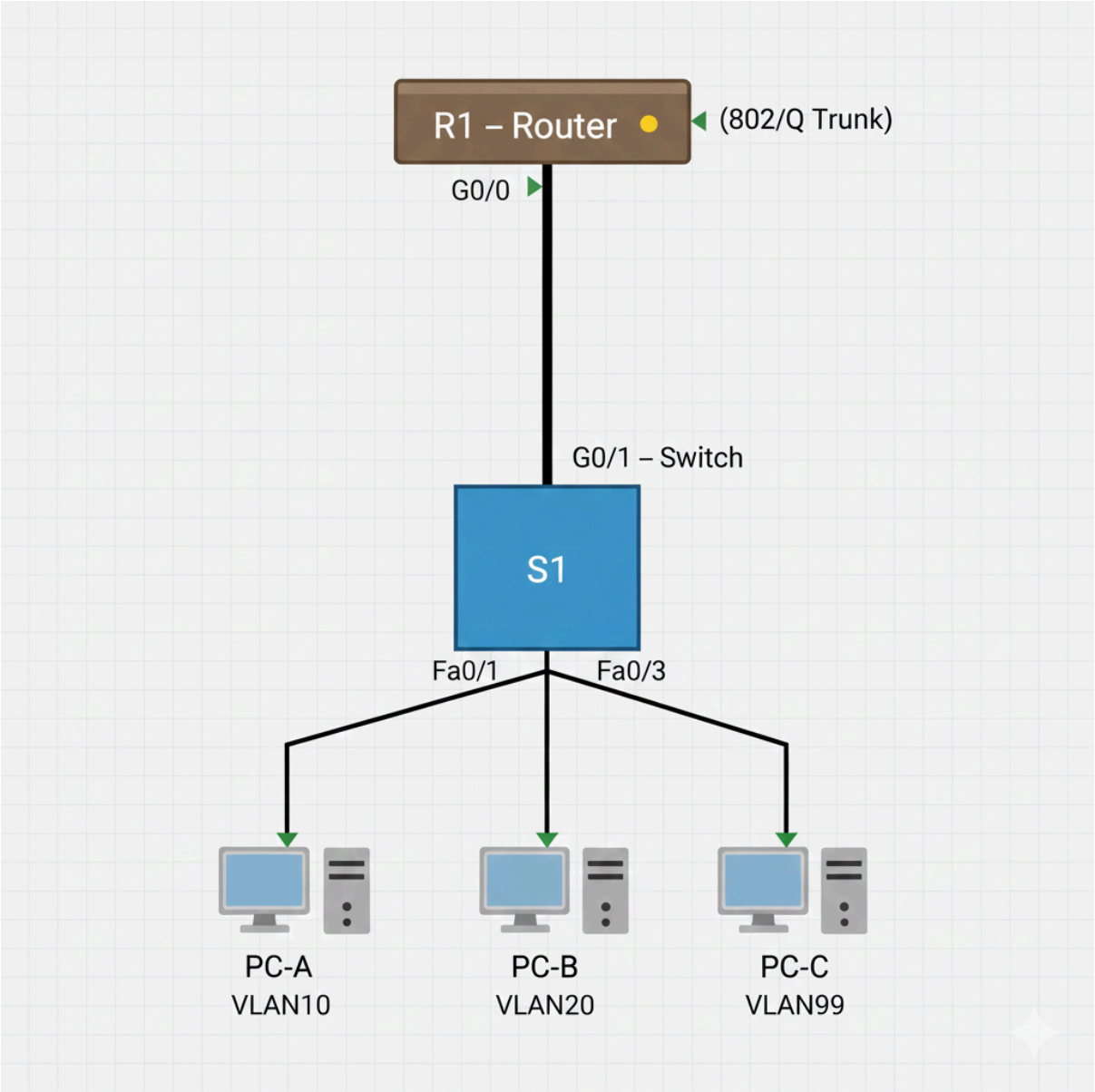
รับผิดชอบ Part 3(Leader)

File:LAP 2

📊 TOPOLOGY DIAGRAM

```
                [ R1 – Router ]
                G0/0 (802.1Q Trunk)
                          |
                          |
                [ G0/1 – Switch ]
                          |
        ┌─────────────────┴─────────────────┐
        |                S1                 |
        |                |                  |
[Fa0/1]                [Fa0/2]            [Fa0/3]
        |                |                  |
        |                |                  |
        [PC-A]  [PC-B]                      [PC-C]
        VLAN10   VLAN20                     VLAN99
```



📋 ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
- **Subnet Mask (/26):** 255.255.255.192
- **Subnets Used:**
  - VLAN 10 → 192.168.10.0/26
  - VLAN 20 → 192.168.10.64/26
  - VLAN 99 → 192.168.10.128/26

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

🎯 LAB OBJECTIVES

By completing this lab, you will be able to:

Part 1: Design and configure multiple VLANs on a switch **Part 2:** Implement Router-on-a-Stick for inter-VLAN routing **Part 3:** Apply subnetting using /26 networks **Part 4:** Isolate management and unused ports for security **Part 5:** Verify and troubleshoot Layer 2 and Layer 3 connectivity

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🔧 LAB INSTRUCTION STEPS

PART 1: SWITCH CONFIGURATION (35 minutes)

Step 1: Basic Switch Setup

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#
```

Step 2: Create VLANs

```
S1(config)# vlan 10
S1(config-vlan)# name USERS
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# name SERVERS
S1(config-vlan)# exit
S1(config)# vlan 99
S1(config-vlan)# name MANAGEMENT
S1(config-vlan)# exit
S1(config)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config-vlan)# exit
```

```
S1(config)#vlan 10
S1(config-vlan)#name USERS
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#exit
S1(config)#vlan 99
S1(config-vlan)#name MANAGEMENT
S1(config-vlan)#exit
S1(config)#vlan 999
S1(config-vlan)#name BLACKHOLE
S1(config-vlan)#exit
S1(config)#
```

Step 3: Secure Unused Ports (Security First)

```
S1(config)# interface range fa0/4-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

```
S1(config)#interface range fa0/4-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 999
S1(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S1(config-if-range)#exit
S1(config)#
```

Step 4: Assign Access Ports

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#interface fa0/3
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 5: Configure Trunk to Router

```
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#interface g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,20,99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 6: Configure Management SVI

```
S1(config)# interface vlan 99
```

```
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# ip default-gateway 192.168.10.129
S1(config)# end
S1# copy running-config startup-config
```

```
S1(config)#interface vlan 99
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

S1(config-if)#ip address 192.168.10.131 255.255.255.192
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#ip default-gateway 192.168.10.129
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

PART 2: ROUTER CONFIGURATION (30 minutes)

Step 7: Basic Router Setup

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Step 8: Enable Physical Interface

```
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#
```

Step 9: Configure Router-on-a-Stick Subinterfaces

```
R1(config)# interface g0/0.10
R1(config-if)# encapsulation dot1Q 10
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.20
R1(config-if)# encapsulation dot1Q 20
R1(config-if)# ip address 192.168.10.65 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.99
R1(config-if)# encapsulation dot1Q 99
R1(config-if)# ip address 192.168.10.129 255.255.255.192
R1(config-if)# exit

R1(config)# end
R1# copy running-config startup-config
```

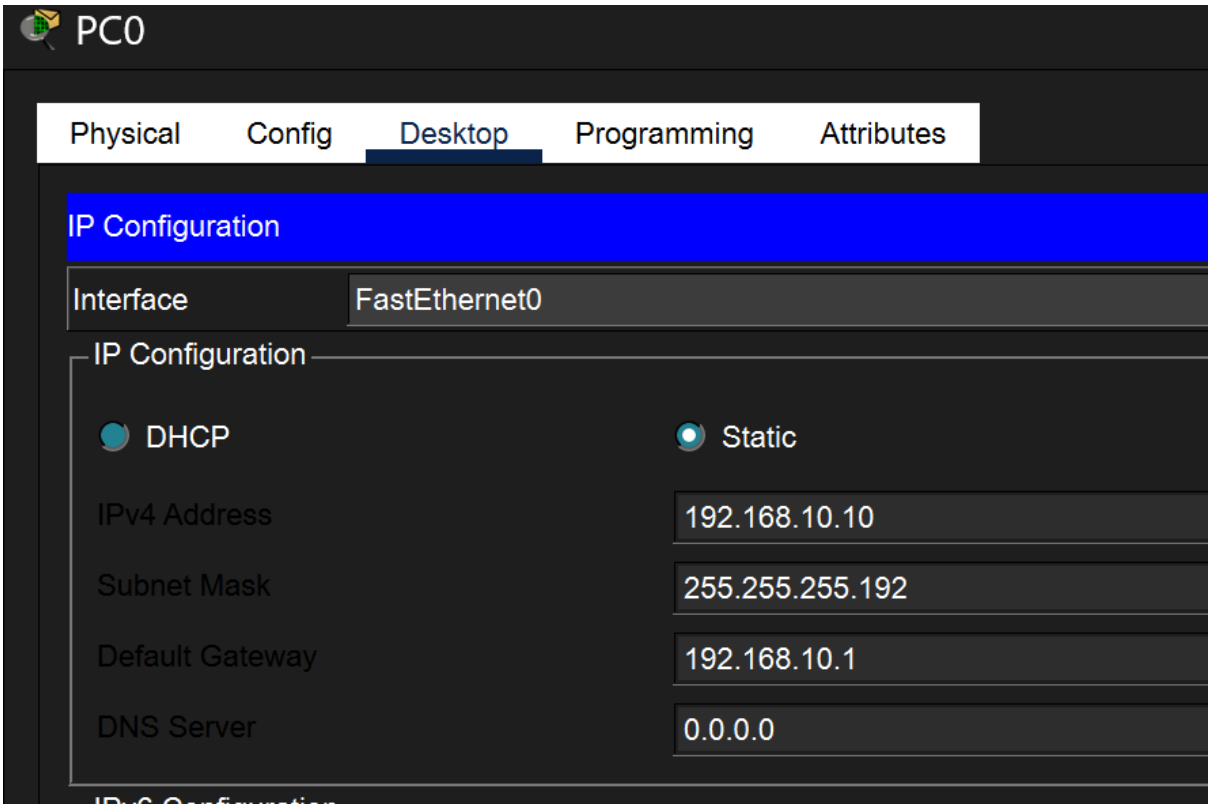
```
R1(config)#interface g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config)#interface g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.10.65 255.255.255.192
R1(config-subif)#exit
R1(config)#interface g0/0.99
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.10.129 255.255.255.192
R1(config-subif)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

PART 3: END DEVICE CONFIGURATION (15 minutes)

Configure each PC manually.

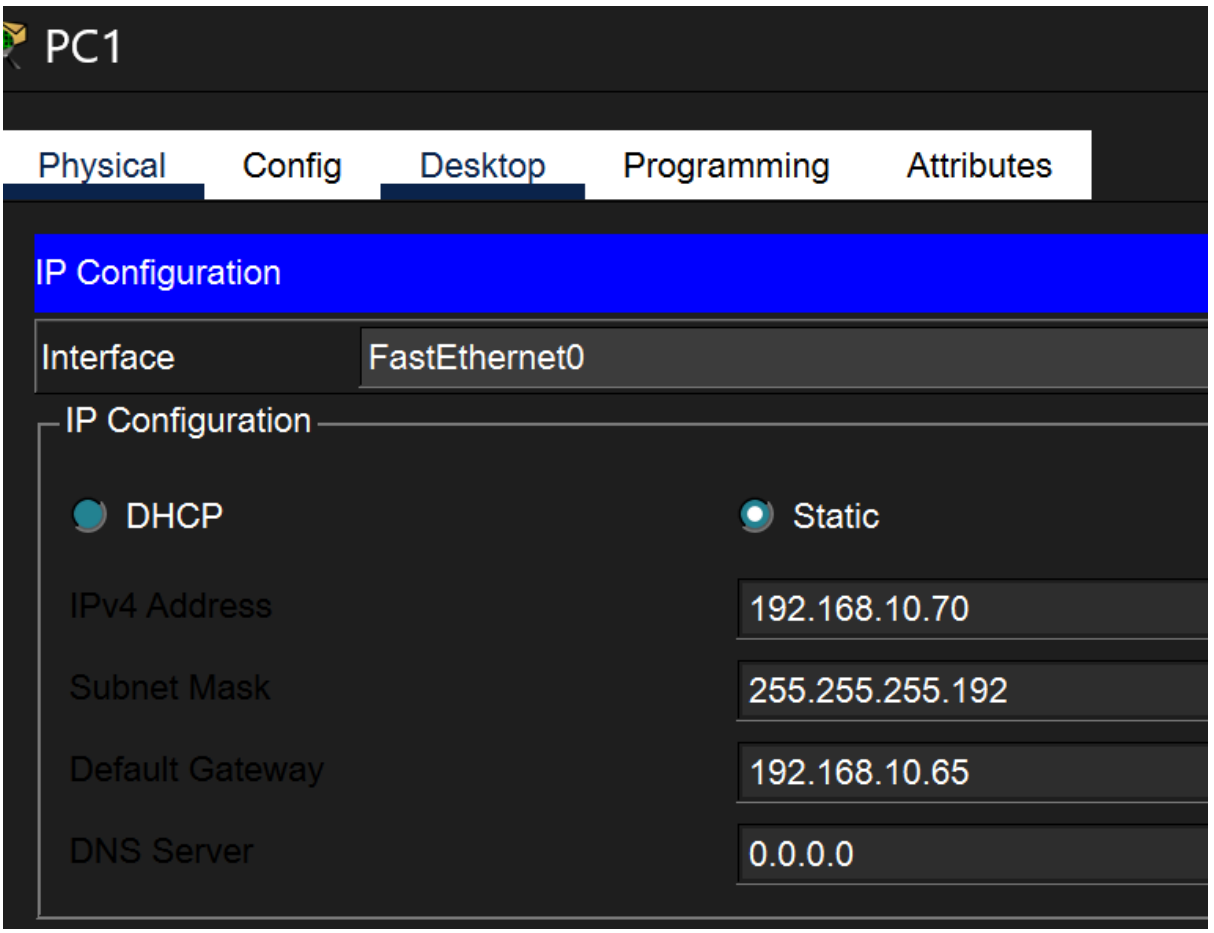
PC-A (VLAN 10)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1



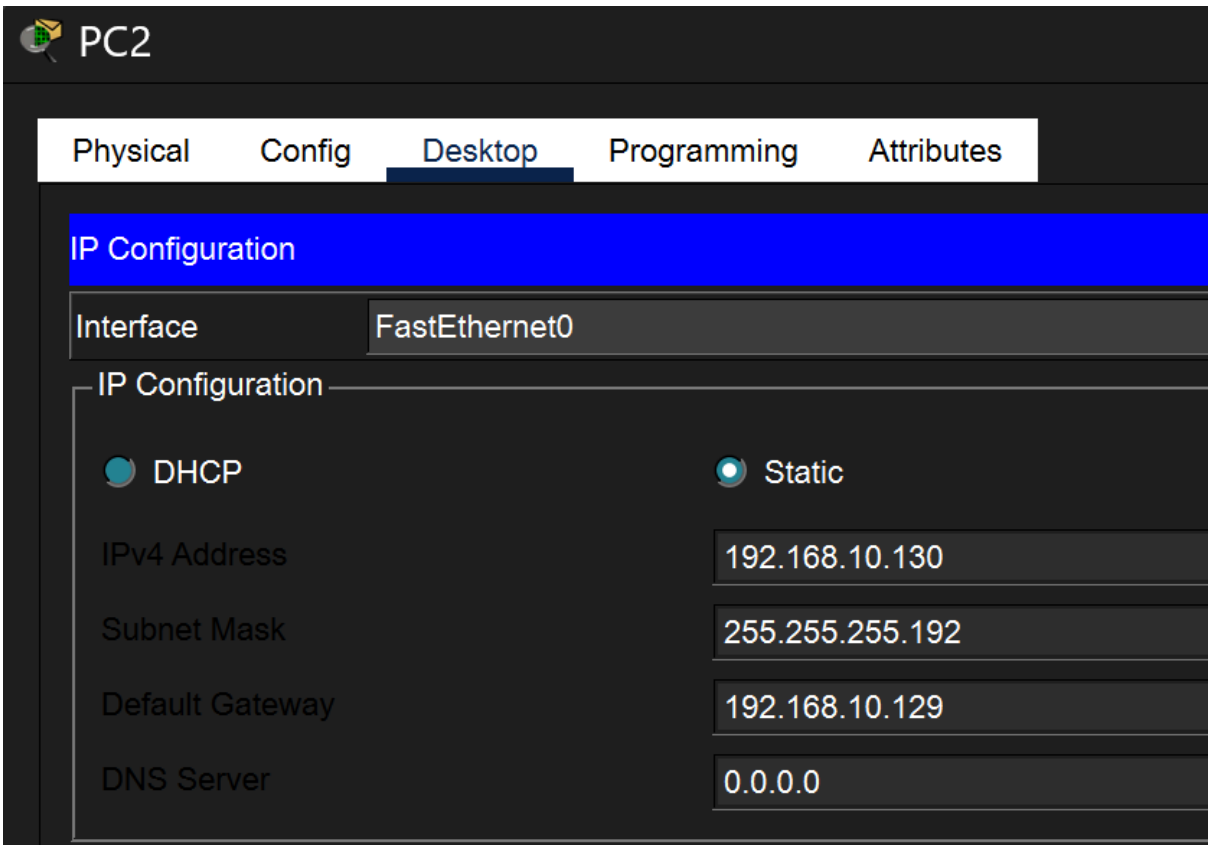
PC-B (VLAN 20)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65



PC-C (VLAN 99)

- IP: 192.168.10.130
- Mask: 255.255.255.192
- Gateway: 192.168.10.129



PART 4: CONNECTIVITY VERIFICATION (20 minutes)

Step 10: Verification Commands

From S1

```
show vlan brief
show interfaces trunk
```

```
S1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99

S1>
```

From R1

```
show ip interface brief
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.10.65   YES manual up                    up
GigabitEthernet0/0.99  192.168.10.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

From PCs

```
ping 192.168.10.1
ping 192.168.10.70
ping 192.168.10.131
```

Expected:

- PC-A ↔ PC-B ✅
- PC-A ↔ S1 Management IP ✅

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10..70
Ping request could not find host 192.168.10..70. Please check the name and try again.
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=22ms TTL=128
Reply from 192.168.10.10: bytes=32 time=3ms TTL=128
Reply from 192.168.10.10: bytes=32 time=6ms TTL=128
Reply from 192.168.10.10: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 22ms, Average = 9ms

C:\>
```

PART 5: TROUBLESHOOTING CHALLENGE (25 minutes)

Scenario A: Wrong VLAN Assignment

- Move PC-B to the wrong VLAN

```
S1>enable
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface fa0/2
S1(config-if)#switchport access vlan 10
S1(config-if)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

- Test connectivity

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Identify failure point
  ตอนนี้อยู่ใน VLAN 10 ใน วง subnet 192.168.10.0/26
Gateway 192.168.10.65 อยู่อีก subnet (192.168.10.64/26)

- Restore correct VLAN

```
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface fa0/2
S1(config-if)#switchport access vlan 20
S1(config-if)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#
```

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time=1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk

```
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface g0/1
S1(config-if)#switchport trunk allowed vlan remove 99
S1(config-if)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#
```

- Test access to S1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>arp -d
C:\>ping 192.168.10.131 -n 20

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time=11ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.131:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 0ms

C:\>
```

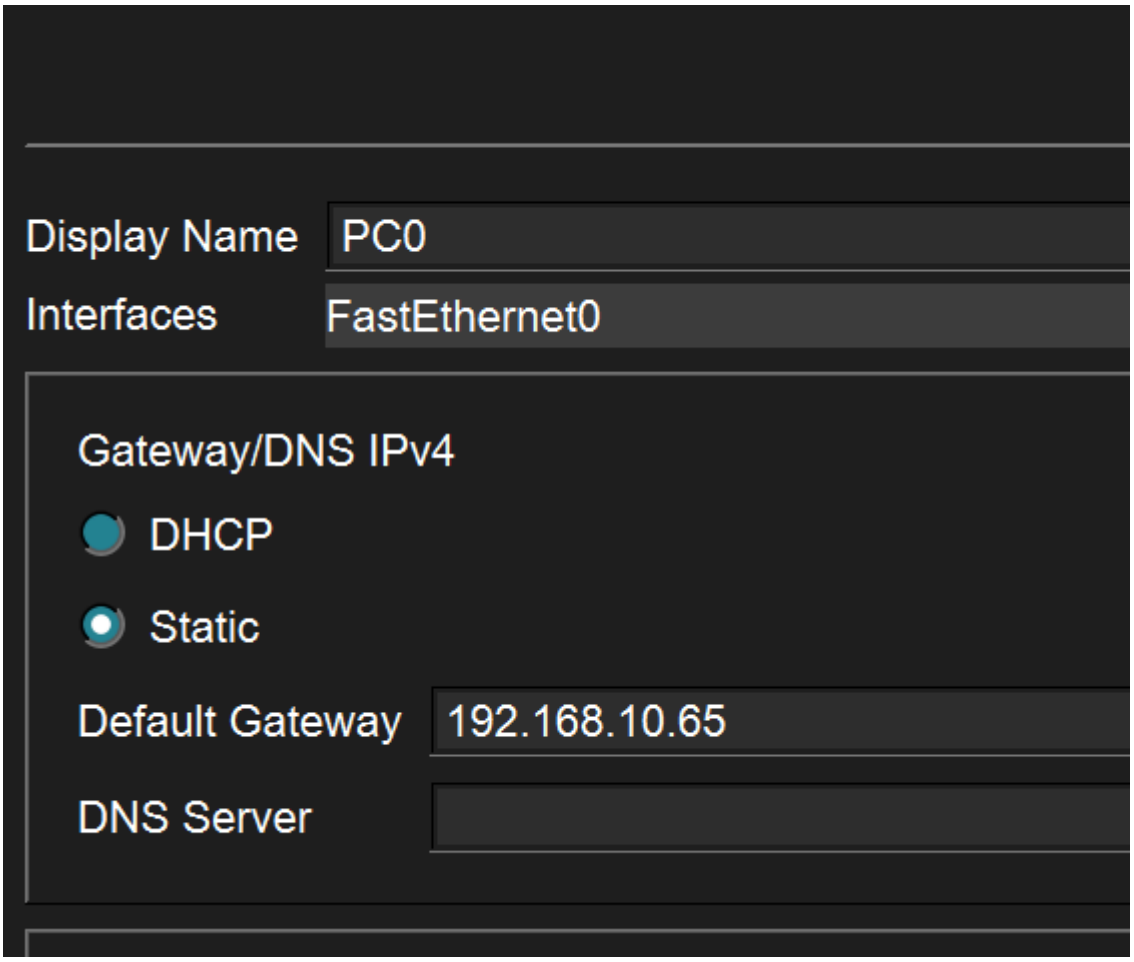
- Explain why management fails
  PC-C → Switch (192.168.10.131):
  Ping 20/20 packets สำเร็จ
  อยู่ VLAN 99 เดียวกัน
  PC-C → Router (192.168.10.129): Request timed out

- Restore configuration

```
S1>enable
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface g0/1
S1(config-if)# switchport trunk allowed vlan add 99
S1(config-if)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#
```

Scenario C: Wrong Gateway

- Change PC-A gateway

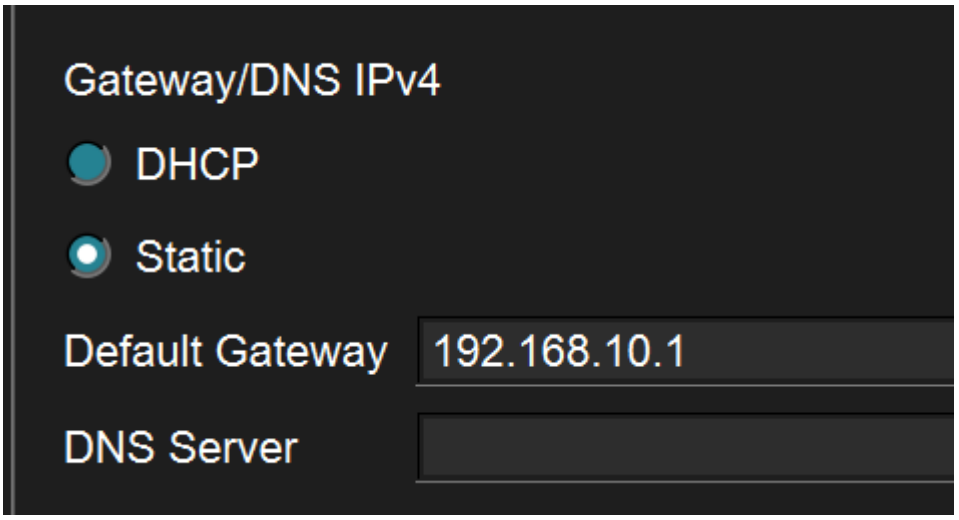


- Observe ping failure

```
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Reply from 192.168.10.129: bytes=32 time=15ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>arp -d
C:\>ping 192.168.70 -n 20
Ping request could not find host 192.168.70. Please check the name and try again.
C:\>ping 192.168.10.70 -n 20

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=12ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=11ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=12ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 1ms
```

- Use `ipconfig` and `ping`



- Fix gateway

```
C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
R1#show interface g0/0.10
GigabitEthernet0/0.10 is up, line protocol is up (connected)
  Hardware is PQUICC_FEC, address is 00d0.d35e.b501 (bia 00d0.d35e.b501)
  Internet address is 192.168.10.1/26
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation 802.1Q Virtual LAN, Vlan ID 10
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last clearing of "show interface" counters never

R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.10.65   YES manual up                    up
GigabitEthernet0/0.99  192.168.10.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

-PC-A ping 192.168.10.65 (Router VLAN20) ได้ PC-A อาจอยู่ใน VLAN 20 โดยไม่ได้ตั้งใจ (ไม่ได้อยู่ใน VLAN 10)
-แม้ผลจะไม่เป็นไปตามทฤษฎี 100%แต่เรียนรู้ได้ว่า:
  -Gateway ต้องอยู่ใน subnet เดียวกัน
  -VLAN assignment สำคัญมาก
  -ต้องตรวจสอบ configuration จริงเสมอ

📝 LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	เจอ 10, 20, 99, 999	pass
Trunk active	show int trunk	VLANs allowed	10, 20, 99 allowe	pass
Router subifs	show ip int br	up/up	up/up	pass
Inter-VLAN ping	ping	Success	Pingเจอ 192.168.10.70	pass

Worksheet 2: Design Understanding

Answer briefly:

1. Why is VLAN 99 used for management?
   …เพราะปลอดภัยกว่า แยกการตั้งค่าออกจากเน็ตที่PCใช้
2. Why is VLAN 999 assigned to unused ports?
   …ใช้ล็อคพอร์ตที่ไม่ใช้ ป้องกันแฮกเกอร์มาต่อสายแอบเข้าเน็ต
3. What does 802.1Q tagging do?
   …การติดป้ายบอกว่าแพ็กเกจนี้อยู่ใน VLAN ไหน

4. Why does each VLAN require a different subnet?
   …แต่ละ VLAN ต้องใช้ subnet ต่างกันเพราะมันเป็น network คนละวงที่แยกจากกัน

🎓 ASSESSMENT QUESTIONS

1. Which device performs inter-VLAN routing?
   …Router (หรือ Layer 3 Switch)
2. Why can PCs in different VLANs communicate?
   …PC ใน VLAN ต่างกันคุยกันได้เพราะมี Router คอยเป็นผู้ส่งข้าม VLAN
3. What breaks if the trunk is removed?
   …ถ้า trunk หาย ทุก VLAN จะคุยกันไม่ได้ เพราะ Router รับ packet ไม่รู้จัก VLAN
4. Why is VLAN 1 avoided?
   …VLAN 1 หลีกเลี่ยงเพราะมันเป็น VLAN เริ่มต้นที่ไม่ปลอดภัยและเป็นเป้าหมายโจมตีของแฮกเกอร์
5. What security risk exists if unused ports remain active?
   …ถ้าพอร์ตที่ไม่ใช้ยังเปิดอยู่ คนนอกสามารถแอบมาต่อสายแล้วแฮกเข้าเน็ตเราได้เลย

🎯 LAB COMPLETION CRITERIA

✔ All VLANs correctly created ✔ Router-on-a-Stick functioning ✔ Management VLAN isolated ✔ Unused ports secured ✔ All troubleshooting scenarios resolved

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